

Assignment 1- Operating System

Maximum Marks: 50
Deadline: Posted on LMS

*Please note that plagiarism will not be tolerated and strict action will be taken against the culprits.
Submit your own work. Good Luck

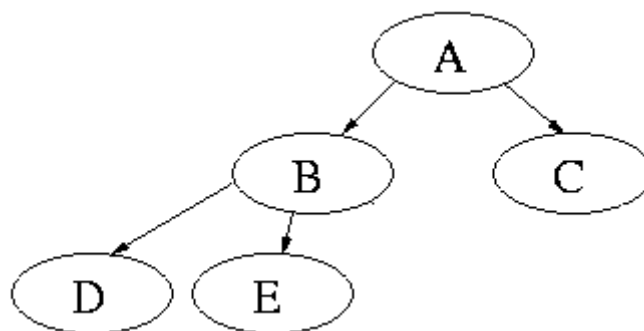
Answer the following questions

1. Difference between Linux and Windows Operating System
2. What are the various functions of Kernel? Explain the services provided by a Kernel?
3. Explain Microkernel and Monolithic Kernel also list key Differences.
4. Write a program in C/C++ to print first 5 odd numbers through child process. parent process must wait for the child process then after 2 seconds delay parent process should display first 5 even numbers.

Sample Output:

```
Odd Numbers - Child Process
1
3
5
7
9
Even Numbers - Parent Process
0
2
4
6
8
```

5. Make the following specific process tree using fork() system call.



Sample Output:

```
Parent of all [A]: 18479
Child [B] with id: 18480 and its Parent id: 18479
Child [c] with id: 18481 and its Parent id: 18479
Child [D] with id: 18482 and its Parent id: 18480
Child [E] with id: 18483 and its Parent id: 18480
```